

Programs Based on Exception Handling in Java

1. Banking Withdrawal System

A banking application should not allow withdrawal if the balance is insufficient. Instead of using predefined exceptions, the system should generate a custom exception.

☞ How will you design and handle a user-defined exception for this case?

✓ 2. Student Age Validation

In an online admission system, students below 18 years are not eligible. The system should throw a custom exception if age is invalid.

☞ How will you implement and use a user-defined exception here?

✓ 3. Employee Salary Check

An organization wants to ensure that employee salary cannot be less than ₹10,000. If entered below this, a custom exception should be raised.

☞ How will you handle this using user-defined exception handling?

✓ 4. Password Authentication System

A login system checks whether the password meets criteria (length, special characters). If invalid, it should throw a custom exception.

☞ How will you design such an exception and handle it?

✓ 5. Product Stock Management

In an e-commerce system, if a user tries to purchase more items than available in stock, the system should throw a custom exception.

☞ How will you implement this scenario?

✓ 6. File Processing System

A program reads a file and processes data. If the file is not found, handle it using predefined exception, but if file content is invalid, throw a custom exception.

☞ How will you combine both types of exceptions?

✔ 7. Result Processing System

A system calculates student results. If marks are negative or greater than 100, a custom exception should be thrown.

☞ How will you validate input using user-defined exception?

✔ 8. ATM PIN Verification

In an ATM system, if a user enters the wrong PIN more than 3 times, a custom exception should be triggered to block the account.

☞ How will you design this logic?

✔ 9. Vehicle Speed Monitoring

A traffic system monitors vehicle speed. If speed exceeds a certain limit, a custom exception should be raised.

☞ How will you implement this using exception handling?

✔ 10. Online Booking System

In a ticket booking system, if seats are already full, a custom exception should be thrown. Also handle input errors like invalid seat number using predefined exceptions.

☞ How will you handle both cases together?